Chapter 1 Introduction

- The vibrating string;
  Solution of the following equation
  $$-kx(t) = m\ddot{x}(t)$$
  This second order ordinary differentail equation becomes
  $$\ddot{x}(t) + c^2 x(t) = 0 \qquad c = \sqrt{k/m}$$
  The general solution is
  $$x(t) = acos(ct) + bsin(ct)$$
  Where a, b are constant give by $x(0)$ and $\dot{x}(0)$.
  Furthermore,
  $$x(t) = \sqrt{a^2 + b^2}cos(ct - \varphi) = Acos(ct - \varphi)$$
- Standing and Traveling Waves;
  - Standing Wave:
    $$...$$
  - Traveling Wave:
    $$...$$
- Harmonic and Superposition of Tones;
-

1.1 Solution to the Wave Equation

1.2 Fourier Coeffiecient

Suppose f is a Riemann integrable functon

$$f : T = \mathbb{R}/2\pi\mathbb{Z} \to \mathbb{C} \text{ or } \mathbb{R}$$

We have

$$\hat{f}(r) = \frac{1}{2\pi}\int_0^{2\pi} f(t)e^{-irt}dt = \frac{1}{2\pi}\int_T f(t)e^{-irt}dt$$

According to the research in 19th,

▶ ***Theorem*** (Fejer's Theorem)

Chapter 2 Fourier Series

2.1 Fundamentals

2.1.1 Fourier Coeffiecient

Definition 2.1.1.1 Define:

$$a_n := \frac{1}{L}\int_0^L f(x)e^{-\frac{2\pi i}{L}x}dx; n \in \mathbb{Z}$$

Remark We now assume that f is Riemann-integrable on $[0, L]$.

2.1.2 Some of Concepts in Analysis

Everywhere Continuous Functions

Piecewise Continuous Functions

2.1.3 Definitions

Definition 2.1.3.1 Take the following notion of

$$\hat{f}(n) := a_n$$

Then, the **Fourier series** is given formally by:

$$\sum_{n=-\infty}^{\infty} \hat{f}(n)e^{\frac{2\pi inx}{L}}$$

Then,

$$f(x) \sim \sum_{n=-\infty}^{\infty} a_n e^{\frac{2\pi inx}{L}}$$

Especially: Suppose f is integrable on $[-\pi, \pi]$, then the nth Fourier coefficient of f is

$$\hat{f}(n) = a_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)e^{-in\theta}d\theta; n \in \mathbb{Z}$$

Thus

2.1.4 Calculating of Some Fourier Coefiecients

Poisson Kernel

$$D_N(x) = \sum_{n=-N}^{N} e^{inx}$$

is called n**th** Dirchlet kernel. It can be seen as the Fourier series with property $a_n = \begin{cases} 1 & |n| \leq N \\ 0 & elsewise \end{cases}$.

A **closed form formula** for Dirchlet kernel is

$$D_N(x) =$$

Dirichlet Kernel

2.1.5 Uniqueness

2.1.6 Convolutions

2.2 Good Kernels

2.3 title

2.4 Convergence

Chapter 3 Fourier Transforms

Chapter 4 Finite Fourier Analysis